

Q1. A system connects 8 registers (16-bit) using a bus with multiplexers. How many multiplexers are required?

- (a) 8
- (b) 16
- (c) 32
- (d) 4

Ans: (b) 16

Q2. A memory unit has 2048 words. How many address lines are needed?

- (a) 10
- (b) 11
- (c) 12
- (d) 13

Ans: (b) 11

Q3. Which bus implementation is more hardware-efficient for large systems?

- (a) Multiplexer
- (b) Tri-state buffer
- (c) Decoder
- (d) Encoder

Ans: (b) Tri-state buffer

Q4. A 4-bit shift register contains 1011. After one left shift with input 0, output is:

- (a) 0110
- (b) 1101
- (c) 0101
- (d) 1110

Ans: (a) 0110

Q5. Number of flip-flops needed for a 64-bit shift register:

- (a) 16
- (b) 32
- (c) 64
- (d) 128

Ans: (c) 64

Q6. In DMA transfer, CPU role is:

- (a) Fully active
- (b) Partially bypassed
- (c) Not used
- (d) Only for control

Ans: (b) Partially bypassed

Q7. In bus transfer $R1 \leftarrow R2$, which happens?

- (a) R1 to bus
- (b) R2 to bus
- (c) Both to bus
- (d) No transfer

Ans: (b) R2 to bus

Q8. To select 8 registers, control lines required:

- (a) 2
- (b) 3
- (c) 4
- (d) 8

Ans: (b) 3

Q9. A right shift register input = 1101, with input 1. Result after 1 shift:

- (a) 1110
- (b) 0110
- (c) 1011
- (d) 1111

Ans: (a) 1110

Q10. First step in memory read operation:

- (a) MDR gets data
- (b) Read signal
- (c) MAR gets address
- (d) Bus transfer

Ans: (c) MAR gets address

Q11. For 16 registers in a bus system, selection bits needed:

- (a) 2
- (b) 3
- (c) 4
- (d) 5

Ans: (c) 4

Q12. Advantage of serial transfer:

- (a) Faster
- (b) More wires
- (c) Less hardware
- (d) Complex

Ans: (c) Less hardware

Q13. Circular left shift of 1001 gives:

- (a) 0011
- (b) 0110
- (c) 1100
- (d) 0001

Ans: (a) 0011

Q14. Arithmetic right shift preserves:

- (a) LSB
- (b) MSB
- (c) Both
- (d) None

Ans: (b) MSB

Q15. Bus with 4 registers requires how many selection lines?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Ans: (b) 2